SE3040 – Application Frameworks
BSc (Hons) in Information Technology
Software Engineering Specialization
3rd Year
Faculty of Computing
SLIIT

2026 - Practical
Lab 3

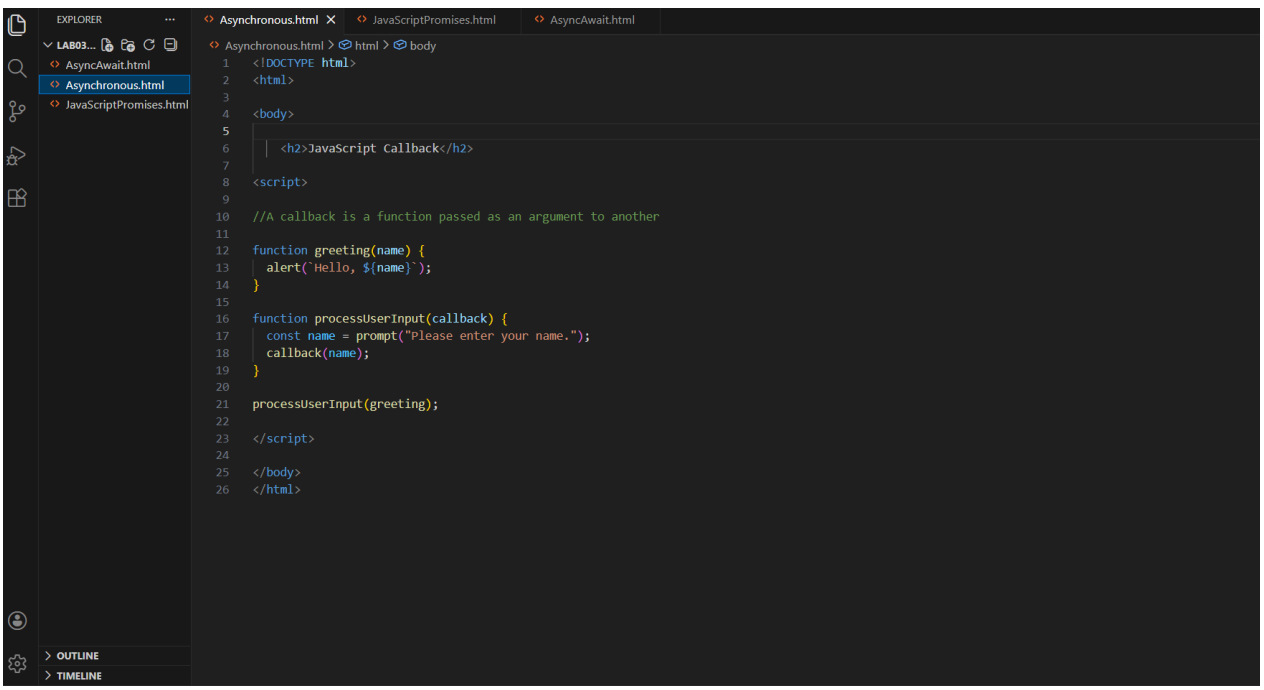
Lab session 3 – Asynchronous JavaScript & basics of Git

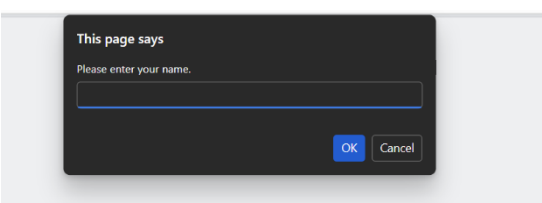
Objective: Teach a set of basic concepts in the JavaScript programming language and Git.

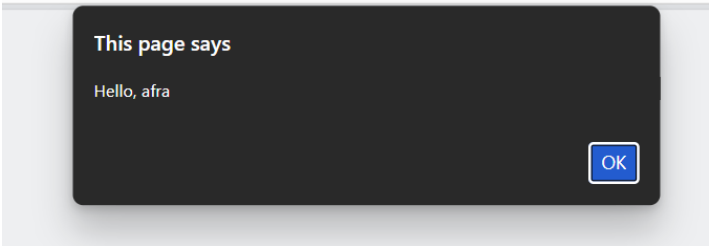
GitHub Link : https://github.com/AfraHameed/Lab03_ApplicationFramework.git

Asynchronous JavaScript

a. JavaScript Callback







b. JavaScript Promises

```html
<html>
<body>

<p id="demo"></p>

<script>
function myMarksDisplayer(some) {
  document.getElementById("demo").innerHTML = some;
}

//A Promise is a JavaScript object that links producing code and
let myPromise = new Promise(function(myResolve, myReject) {
  let marks = 60;

// some code (try to change marks to different numbers)

  if (marks <= 100 && marks >=0) {
    myResolve("Your marks is : " + marks);
  } else {
    myReject("Error");
  }
});

myPromise.then(
  function(value) {myMarksDisplayer(value);},
  function(error) {myMarksDisplayer(error);}
);
</script>

</body>
</html>
```

JavaScript Promise

Your marks is : 60

c. JavaScript Async & Await

```html
<!DOCTYPE html>
<html>
<body>

<h2>JavaScript Async & Await</h2>

<p id="demo"></p>

<script>

//async makes a function return a Promise
//await makes a function wait for a Promise

function resolveAfter2Seconds() {
    return new Promise(resolve => {
      setTimeout(() => {
        resolve('resolved - success');
      }, 2000);
    });
  }

  async function asyncCall() {
    console.log('calling');
    const result = await resolveAfter2Seconds();
    console.log(result);
    // Expected output: "resolved"
  }

  asyncCall();

</script>

</body>
</html>
```

```
calling
resolved - success
>
```

Git

```
Microsoft Windows [Version 10.0.26100.7623]
(c) Microsoft Corporation. All rights reserved.

C:\Users\ksaranya\OneDrive\Desktop\lab3AF>git log
commit 981cb26fd27594c8fa1993be793b3db757048082 (HEAD -> main, origin/main)
Author: SK <kirupananthansaranya@gmail.com>
Date:   Fri Feb 13 13:55:34 2026 +0530

    Lab 3 - Async JS tasks completed

C:\Users\ksaranya\OneDrive\Desktop\lab3AF>
```